

Suggested Teaching Guideline for
Security Concepts
PGCP-ITISS February 2026

Duration: 60 Theory hrs + 50 lab hrs+ 10 hrs Self Learning (Total: 120 hrs)

Objective: To introduce the student to issues related to application security and ethical Hacking.

Prerequisites: Knowledge of OS, Networking, programming and database concepts

Evaluation method: Theory exam – 40% weightage
Lab exam – 40% weightage
Internal Assessment– 20% weightage

List of Books / Other training material

Courseware: Gray Hat Hacking: The Ethical Hackers Handbook by Shon Harris, 5th Ed

Reference: Python for Unix and Linux System Administration

Application Security
(Web Application Security): (14T+ 14L +4 SL)

Session 1: (2T + 3L)

Theory:

- OWASP Top 10 – 2021
- Injection and Inclusion
- Cross Site Scripting
- Injection in stored procedures

Lab Assignment:

- Give SQL queries to bypass authentication.
- Make your own cross site script and apply in any web application.

Session 2:(2T+2L)

Theory:

- Denial of Service
- Buffer Overflows and Input Validation
- Access Control

Lab Assignment:

- Try DOS attack to denial a service of any server.
- Try to do input validation to get access in vulnerable application.

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Session 3:(3T+3L)**Theory:**

- Web Application Security Risks
- Identifying the Application Security Risks
- Threat Risk Modelling
- Other HTTP fields
- Overview of Burp Suite features

Lab Assignment:

- Identify all risks and vulnerabilities of web applications using tools.
- Configuring Burp Suite with a web browser
- Scanning web applications for vulnerabilities using Burp Suite
- Using Burp Suite's scanner to identify common security issues

Session 4: (3T+2L)**Theory:**

- Data Extraction
- Advanced Identification/Exploitation

Lab Assignment: (2L)

- Find vulnerability of data extraction/exploitation of a web application.
- Find vulnerabilities of HTTP Methods (1.0 & 1.1).
- Analyse web application with the help of Wireshark.

Session 5: (4T+4L+ 4SL)**Theory:**

- SAST and DAST tools
- Case Study On Web Application Framework
- Use browser-jsguard Firefox add-on also to detect Malicious and Suspicious Webpages.

Lab Assignment: (4L)

- Try to do static and Dynamic application security testing by SAST and DAST tools.

Self-Learning: (SL based on session 1-5)

- SQL Injection Assignment: Try to create SQL injection payloads to bypass login forms and other user input fields. Focus on exploring different techniques like time-based blind injection, error-based injection, and union-based injection.
- Cross-Site Scripting (XSS) Assignment: Write scripts to exploit reflected and stored XSS vulnerabilities. Test these scripts in an isolated environment like DVWA (Damn Vulnerable Web Application) or WebGoat.
- Explore and study OWASP's Web Application Security Testing Framework (WSTF). Familiarize yourself with various methodologies to test security in web applications, focusing on attack vectors and

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mitigation strategies.

- Review Common Vulnerabilities and Exposures (CVE) and the National Vulnerability Database (NVD) for insights on emerging vulnerabilities.
- Focus on vulnerabilities related to SQL injection, XSS, and buffer overflows. Look into their historical impact and how they've evolved in different web technologies.
- Case study on SQL Injection: Study real-world SQL injection attacks, such as the Heartland Payment Systems breach. Understand how the attack worked and the impact it had on their systems and data.
- Burp suite exploration: Watch videos or go through documentation on advanced Burp Suite functionalities, such as Intruder, Repeater, and Scanner.
- Research the use of SAST (Static Application Security Testing) and DAST (Dynamic Application Security Testing) tools, such as Checkmarx, Fortify, and OWASP ZAP. Explore how these tools automate the detection of security vulnerabilities in source code and during runtime.

Ethical Hacking (32 T+ 26 L+ 3 SL)

Session 6: (2T)

Theory:

- Security Management Concepts & Principles
- Human side of Information Security's
- Threats of Information System
- Threats and attacks
- Classification of Threads and attacks

Session 7: (2T)

Theory:

- Protecting Information System Security
- Security in Mobile and Wireless Computing
- Credit card frauds in mobile and wireless Computing
- Information Security Management
- Fundamentals of Information Security

Session 8: (2T)

Theory:

- Cyber Crimes
- Understanding Cyber Crimes in context of Internet
- Legal aspects of open communications
- Indian Penal Law & Cyber Crimes Fraud Hacking Mischief
- International law
- Obscenity and Pornography Internet
- Introduction to Ethical Hacking
- Understanding Ethical Hacking Terminology
- Identifying Different Types of Hacking Technologies
- Understanding the Different Phase Involved in Ethical Hacking

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Session 9: (2T)**Theory:**

- Types of Hacker Classes
- Red team , Blue Team ,Grey Team
- Ethical Hackers and Crackers
- Goals of Attackers
- Security, Functionality and Ease of Use Triangle
- Defining the Skills Required to become an Ethical Hacker

Session 10: (2T + 3L)**Theory:**

- Introduction to Ethical Hacking
- Creating a Security Evaluation Plan
- Types of Ethical Hacks
- Foot-printing and Social Engineering
- Understand How Traceroute Is Used in Foot-printing
- Define the Terms Port Scanning, Network Scanning and Vulnerability Scanning
- Understand various Scanning Methodologies
- SYN, Stealth, XMAS, NULL, IDLE and FIN Scans

Lab Assignment: (Note: Do not target Govt sites, Bank sites)

- Use NMAP, WHOIS for Reconnaissance.
- Open-source Intelligence Gathering tool:
 - ◆ Netcraft, Shadon, Google Dorks, Recon-Ng
- Use traceroute in footprinting.
- Use FOCA for finger printing.
- Use Tree Pad tool for reporting
- Using Burp Suite for Reconnaissance:
 - ◆ Spidering the web application
 - ◆ Mapping the attack surface using Burp Suite

Session 11: (2T+ 2L)**Theory:**

- TCP Communication Flag Types
- Banner Grabbing and OS Fingerprinting Techniques
- How Proxy servers are used in launching an Attack
- Http tunneling Techniques
- IP Spoofing Techniques
- Enumeration
- Password-cracking Techniques
- Cracking Windows Passwords
- Redirecting the SMB Logon to the attackers
- SMB Redirection, SMB Relay MITM Attacks and Countermeasures
- NetBIOS DOS Attacks
- DDos Attack

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Lab Assignment:

- Do all these practical of following;
 - ◆ Ip spoofing
 - ◆ DNS Reconnaissance
 - ◆ DNS Enumeration
 - ◆ Host and Port Discovery (using NMAP) etc.
- Use Nessus also to find Vulnerability.

Session 12: (2T+ 2L)**Theory:**

- Password-Cracking Countermeasures
- Active/Passive online Attacks
- Offline Attacks
- Keyloggers and other Spyware Technologies
- Trojans and Backdoors
- Overt and Covert Channels
- Types of Trojans
- Reverse-connecting Trojans
- Netcat Trojan
- Indications of a Trojan Attacks

Lab Assignment:

- Use KALI operating system tools to crack password

Session 13: (2T+2L)**Theory & Lab:**

- Wrapping
- Trojan Construction Kit and Trojan Makers
- The countermeasure Techniques in Preventing Trojans
- Trojan-Evading techniques
- System File Verification

Session 14: (2T+2L)**Theory & Lab:**

- Difference between a Virus and a Worm
- Types of Viruses
- Antivirus Evasion Techniques
- Virus Detection Methods

Session 15: (2T+2L)**Theory & Lab:**

- Protocols Susceptible to Sniffing
- Active and Passive Sniffing
- ARP Poisoning
- Ethereal Capture and Display Filters
- MAC Flooding

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- DNS Spoofing Techniques
- DNS hacking
- Describe Sniffing Countermeasures

Session 16: (2T+2L)

Theory & Lab:

- Types of DOS Attacks
- How DDos Attacks Work
- How BOTs/BOTNETs work
- Smurf Attacks
- SYN Flooding
- Spoofing vs Hijacking
- Types of Session Hijacking
- Steps to perform session Hijacking
- Prevention of session Hijacking

Session 17: (4T+2L)

Theory & Lab:

- Hacking Web Servers
- Web Application Vulnerabilities
- Web-Based Password Cracking Techniques
- Wireless Hacking
- WEP, WPA Authentication Mechanisms and Cracking Techniques
- Wireless Sniffers and Locating SSIDS, MAC spoofing
- Wireless hacking Techniques
- Methods used to secure Wireless Networks

Session 18: (2T + 3L)

Theory:

- Backdoor Devices
- Distributed Dos attacks
- Biometric Spoofing
- Linux Hacking
- Linux Backdoors
- IDSs, Honeypots and Firewalls

Lab Assignment:

- Use NAGIOS, SNORT as IDS/IPS
- Try to install DMZ in your infrastructure.

Session 19: (2T + 3L)

Theory:

- Physical Security
- Overview of Physical Security
- Need of Physical Security
- Factors Affecting Physical Security
- Penetration Testing Methodologies

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Lab Assignment:

- Use Metasploit framework for penetration testing.

Session 20: Malware Reverse Engineering (2T + 3L + 3SL)

Theory:

- Types of Malware
- Malicious code Families
- Latest Trends in Malware
- Analysis of Malware

Lab Assignment:

- Do analysis of a Malware use tools of Static and Dynamic analysis.
- Case Study
- Malware Analysis Scenario – Case Study

Note: We can use **USB Pratirodh** and **AppSamvid** tools for System security.

Self-Learning: (SL based on Session 6-20)

- Practice using NMAP for port scanning, network discovery, and vulnerability scanning.
- Perform WHOIS and DNS reconnaissance on a public domain, gathering information about the target.
- Use tools like John the Ripper or Hydra to attempt cracking a weak password using dictionary attacks or brute-force techniques.
- Try out some offline cracking using Rainbow Tables or Hashcat for understanding how attackers exploit weak passwords.
- Study the differences between viruses, worms, and Trojans. Focus on how these types of malware spread and the impact they have on systems.
- Dive deeper into Netcat Trojans and Reverse-connecting Trojans. Explore how these backdoors allow attackers to maintain control over infected systems.
- Study a real-world DDoS attack such as the Mirai Botnet attack on Dyn. Understand the attack vectors, the role of botnets, and how smurf attacks and SYN flooding contribute to the attack.
- Familiarize yourself with Penetration Testing Methodologies (such as the PTES). Study how a typical penetration test is structured, focusing on the reconnaissance, scanning, and exploitation phases

Mobile Security: (14T+ 10L+ 3SL)

Session 21: (3T + 2L)

Theory:

- Introduction to Android Architecture
- Android File Structure

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- Android Build Process
- Android App fundamentals
- Android Security Model
- Device Rooting

Lab Assignment

- Android Emulator setup

Session 22: (3T + 2L)

Theory:

- Android Debug bridge
- Penetration Testing Tools

Lab Assignment

- Lab based on Android Debug Bridge

Session 23: (2T + 2L)

Theory:

- OWASP Top 10 Mobile App vulnerabilities
- Attacks on Android Apps

Lab Assignment

- Lab based on Reverse Engineering
- Lab based on Smishing attack

Session 24: (3T + 2L)

Theory:

- Web based attacks on Android devices
- Networks based attacks
- Social Engineering attacks

Lab Assignment

- Lab based on MITM Attacks
- Packet Sniffing using Wireshark
- Lab based on Phishing Attack

Session 25: (3T + 2L + 3SL)

Theory:

- Overview of Mobile Malware
- Android App Analysis

Lab Assignment

- Static and Dynamic Analysis of Android Apps using MobSF

Self-Learning: (SL based on Session 21-25)

- Research the Android Security Model and the risks associated with device rooting. Study how rooting compromises Android security and the

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techniques used by malware to exploit rooted devices.

- Review the OWASP Top 10 Mobile App vulnerabilities. Understand each of the common security flaws in mobile applications, such as Insecure Data Storage, Improper Platform Usage, and Insecure Communication.
- Study the tools and techniques for penetration testing mobile applications. Explore the use of Android Debug Bridge (ADB) and Frida for hooking and inspecting Android app behavior.
- Understand the basics of reverse engineering Android apps, including techniques like decompiling APK files using APKTool and JADX.
- Explore smishing attacks and how they are used to target Android users via SMS and social engineering tactics.
- Study malware in the Android ecosystem. Look into real-world Android malware like Joker or EventBot. Understand how these malware types exploit vulnerabilities to steal sensitive data.
- Research phishing attacks targeting Android users, focusing on tactics used to trick users into revealing their credentials.